

Assignment 2

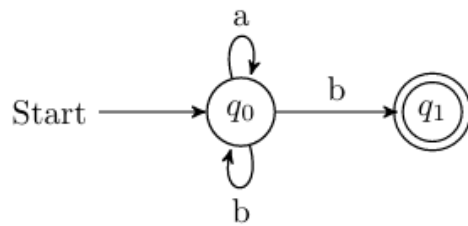
Name:

1 Required Exercises

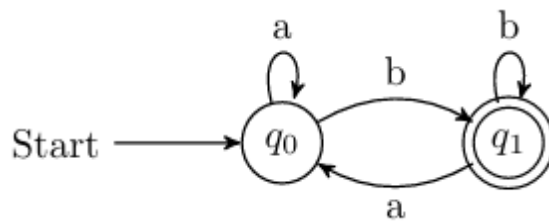
Exercise 1

1. $L((a|b)^*b)$

(1) NFA

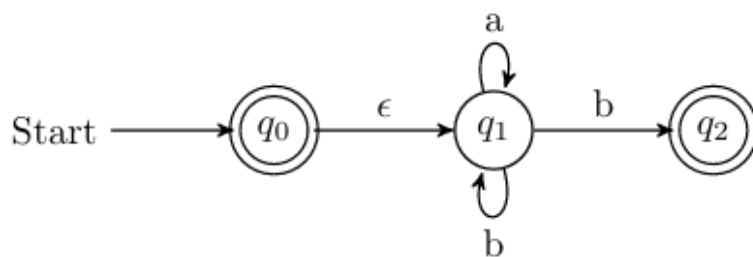


(2) DFA

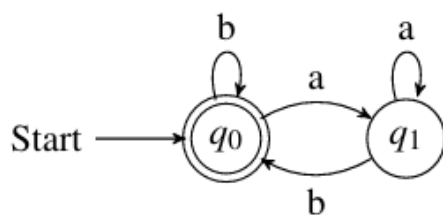


2. $L(((\epsilon|a)^*b)^*)$

(1) NFA

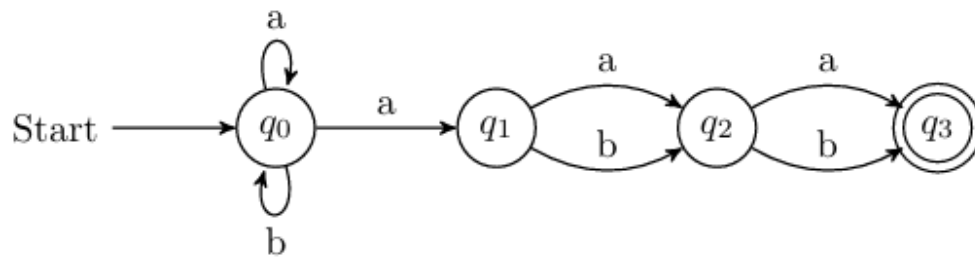


(2) DFA

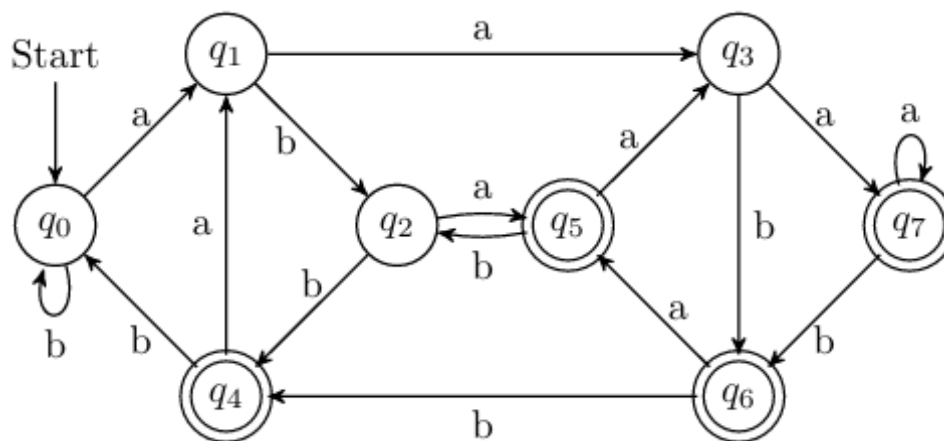


3. $L((a|b)^*a(a|b)(a|b))$

(1) NFA

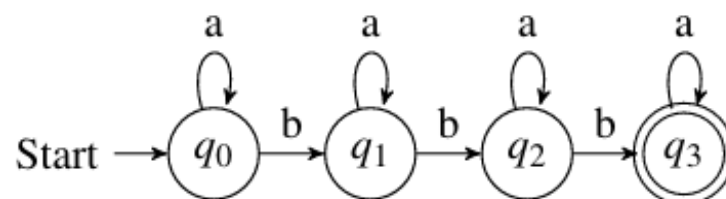


(2) DFA

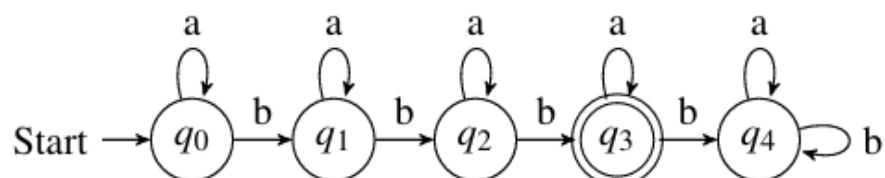


4. $L(a^*ba^*ba^*ba^*)$

(1) NFA



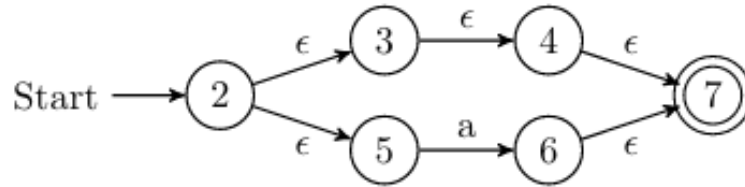
(2) DFA



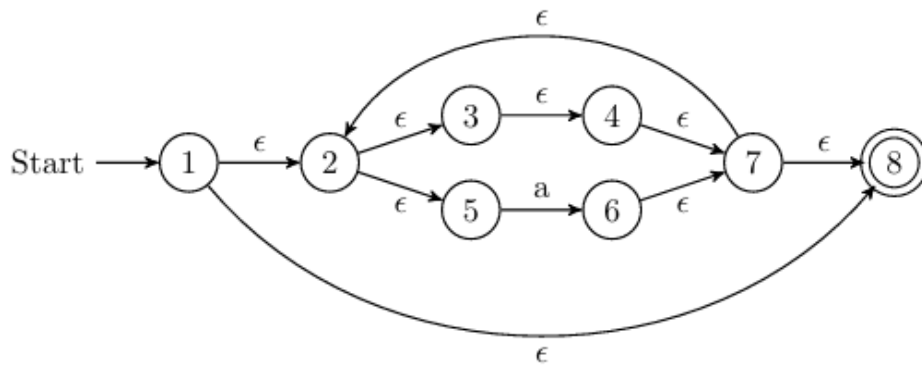
Exercise 2

1. $((\epsilon|a)^*b)^*$

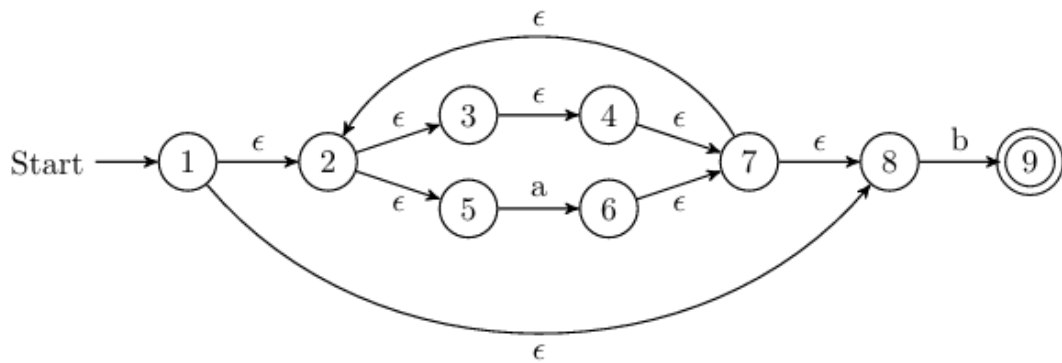
(1) Construct the NFA for $R_1 = \epsilon|a$



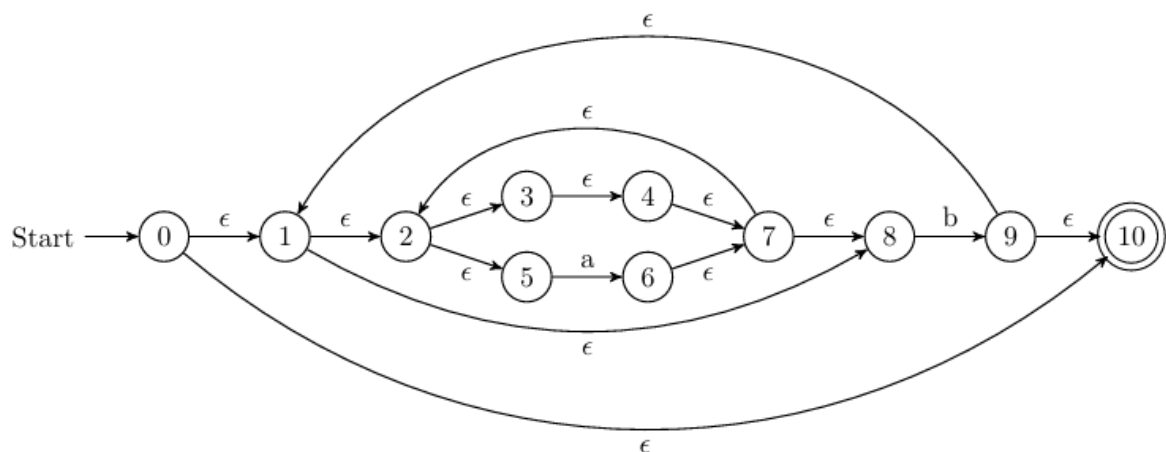
(2) Construct the NFA for $R_2 = (\epsilon|a)^* = R_1^*$



(3) Construct the NFA for $R_3 = (\epsilon|a)^*b = R_2b$

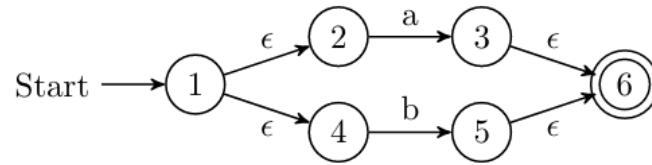


(4) Construct the NFA for $((\epsilon|a)^*b)^* = R_3^*$

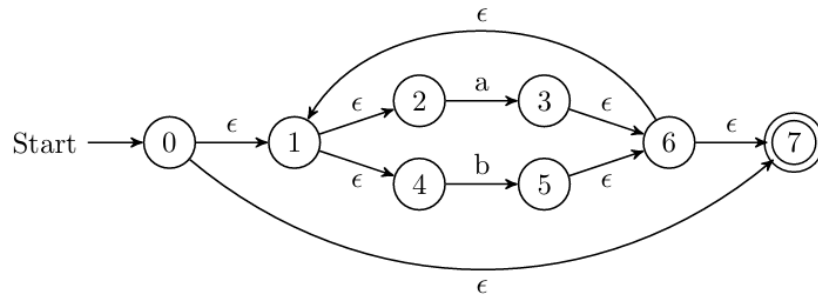


2. $(a|b)^*a(a|b)(a|b)$

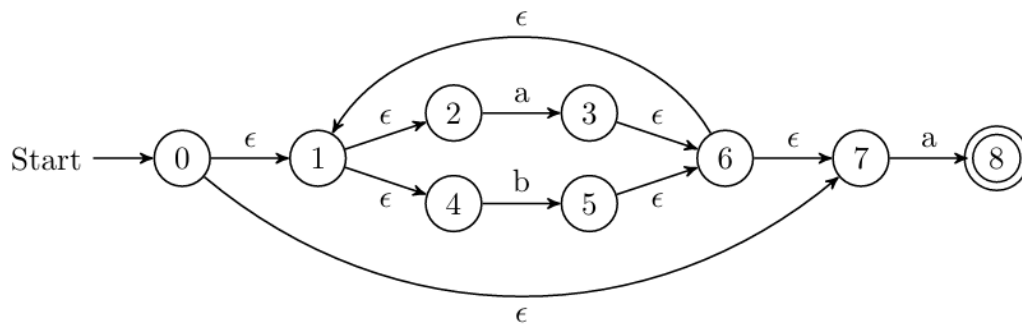
(1) Construct the NFA for $R_1 = a|b$.



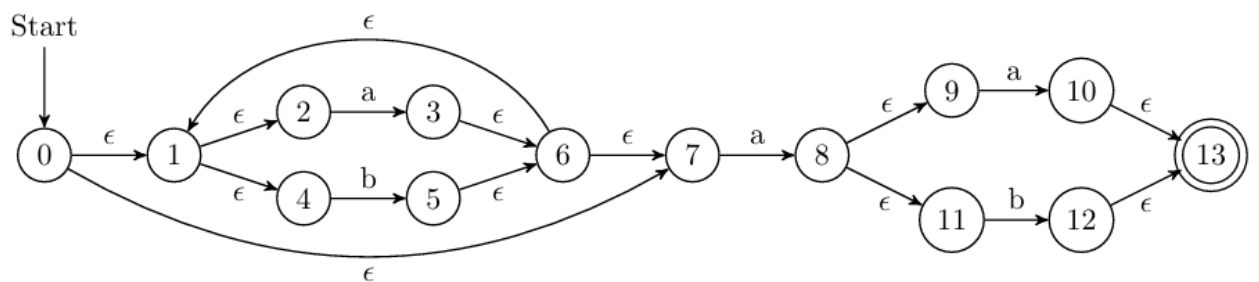
(2) Construct the NFA for $R_2 = (a|b)^* = R_1^*$.



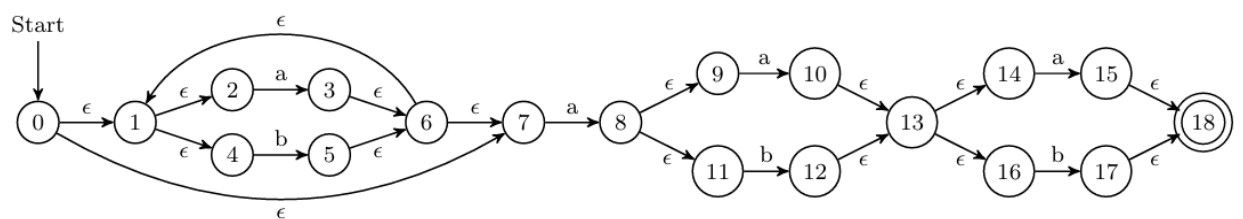
(3) Construct the NFA for $R_3 = (a|b)^*a = R_2a$.



(4) Construct the NFA for $R_4 = (a|b)^*a(a|b) = R_3R_1$.

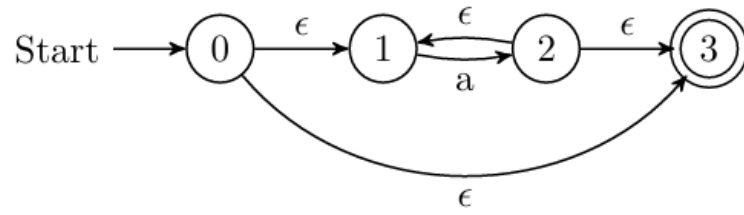


(5) Construct the NFA for $(a|b)^*a(a|b)(a|b) = R_4R_1$.

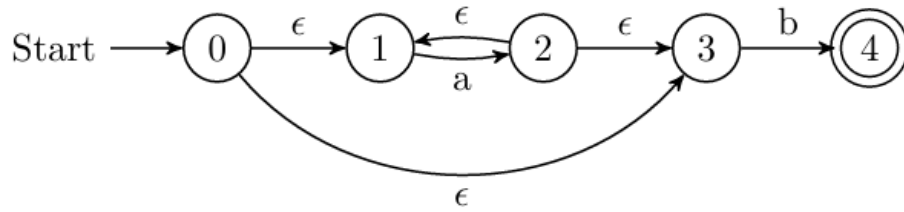


3. $a^*ba^*ba^*ba^*$

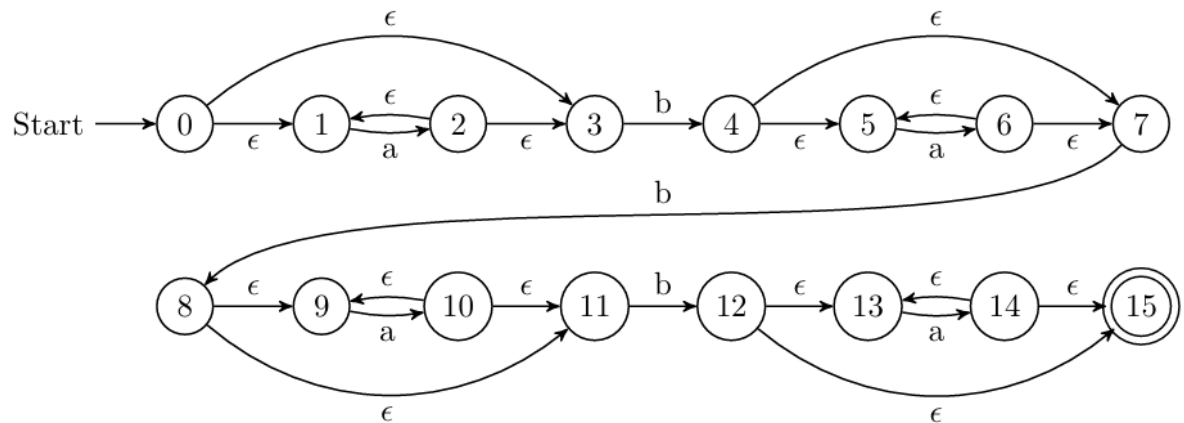
(1) Construct the NFA for $R_1 = a^*$.



(2) Construct the NFA for $R_2 = a^*b = R_1b$.



(3) Construct the NFA for $a^*ba^*ba^*ba^* = R_2R_2R_2R_1$.



Exercise 3

$$1. A = \varepsilon - \text{closure}(\{0\}) = \{0, 1, 2, 3, 4, 5, 7, 8, 10\}.$$

$$B = \delta_D(A, a) = \varepsilon - \text{closure}(\{6\}) = \{2, 3, 4, 5, 6, 7, 8\}.$$

$$C = \delta_D(A, b) = \varepsilon - \text{closure}(\{9\}) = \{1, 2, 3, 4, 5, 7, 8, 9, 10\}.$$

$$D = \delta_D(B, a) = \varepsilon - \text{closure}(\{6\}) = B.$$

$$E = \delta_D(B, b) = \varepsilon - \text{closure}(\{9\}) = C.$$

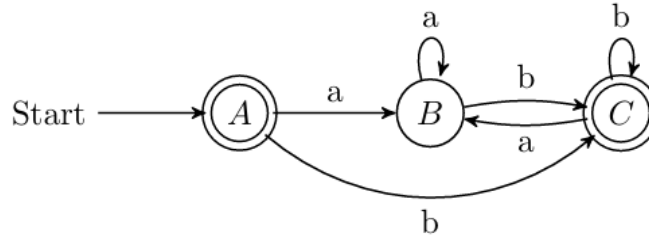
$$F = \delta_D(C, a) = \varepsilon - \text{closure}(\{6\}) = B.$$

$$G = \delta_D(C, b) = \varepsilon - \text{closure}(\{9\}) = C.$$

The transition table of the DFA (Starting state is A and finite states are {A, C}):

NFA STATE	DFA STATE	a	b
{0, 1, 2, 3, 4, 5, 7, 8, 10}	A	B	C
{2, 3, 4, 5, 6, 7, 8}	B	B	C
{1, 2, 3, 4, 5, 7, 8, 9, 10}	C	B	C

The transition diagram:



$$2. A = \varepsilon - \text{closure}(\{0\}) = \{0, 1, 2, 4, 7\}.$$

$$B = \delta_D(A, a) = \varepsilon - \text{closure}(\{3, 8\}) = \{1, 2, 3, 4, 6, 7, 8, 9, 11\}.$$

$$C = \delta_D(A, b) = \varepsilon - \text{closure}(\{5\}) = \{1, 2, 4, 5, 6, 7\}.$$

$$D = \delta_D(B, a) = \varepsilon - \text{closure}(\{3, 8, 10\}) = \{1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13, 14, 16\}.$$

$$E = \delta_D(B, b) = \varepsilon - \text{closure}(\{5, 12\}) = \{1, 2, 4, 5, 6, 7, 12, 13, 14, 16\}.$$

$$F = \delta_D(C, a) = \varepsilon - \text{closure}(\{3, 8\}) = B.$$

$$G = \delta_D(C, b) = \varepsilon - \text{closure}(\{5\}) = C.$$

$$H = \delta_D(D, a) = \varepsilon - \text{closure}(\{3, 8, 10, 15\}) = \{1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 18\}.$$

$$I = \delta_D(D, b) = \varepsilon - \text{closure}(\{5, 12, 17\}) = \{1, 2, 4, 5, 6, 7, 12, 13, 14, 16, 17, 18\}.$$

$$J = \delta_D(E, a) = \varepsilon - \text{closure}(\{3, 8, 15\}) = \{1, 2, 3, 4, 6, 7, 8, 9, 11, 15, 18\}.$$

$$K = \delta_D(E, b) = \varepsilon - \text{closure}(\{5, 17\}) = \{1, 2, 4, 5, 6, 7, 17, 18\}.$$

$$L = \delta_D(H, a) = \varepsilon - \text{closure}(\{3, 8, 10, 15\}) = H.$$

$$M = \delta_D(H, b) = \varepsilon - \text{closure}(\{5, 12, 17\}) = I.$$

$$N = \delta_D(I, a) = \varepsilon - \text{closure}(\{3, 8, 15\}) = J.$$

$$O = \delta_D(I, b) = \varepsilon - \text{closure}(\{5, 17\}) = K.$$

$$P = \delta_D(J, a) = \varepsilon - \text{closure}(\{3, 8, 10\}) = D.$$

$$Q = \delta_D(J, b) = \varepsilon - \text{closure}(\{5, 12\}) = E.$$

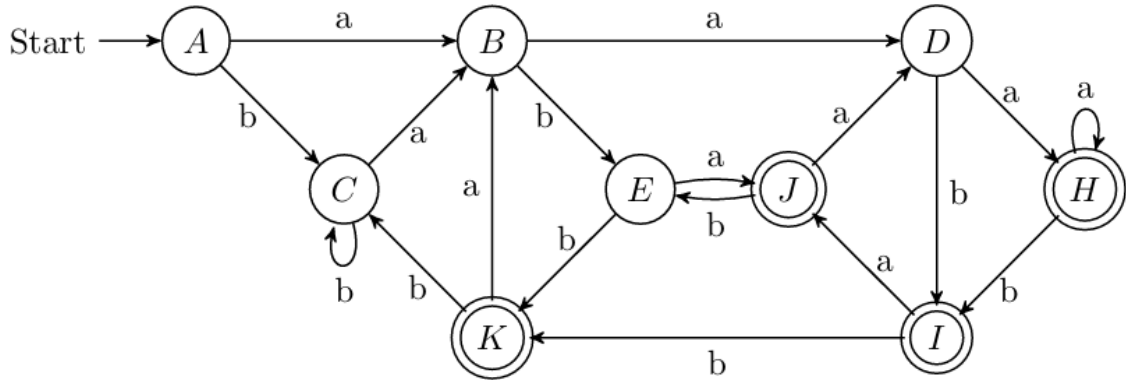
$$R = \delta_D(K, a) = \varepsilon - \text{closure}(\{3, 8\}) = B.$$

$$S = \delta_D(K, b) = \varepsilon - \text{closure}(\{5\}) = C.$$

The transition table of the DFA (Starting state is A and finite states are {H, I, J, K}):

NFA STATE	DFA STATE	a	b
{0, 1, 2, 4, 7}	A	B	C
{1, 2, 3, 4, 6, 7, 8, 9, 11}	B	D	E
{1, 2, 4, 5, 6, 7}	C	B	C
{1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13, 14, 16}	D	H	I
{1, 2, 4, 5, 6, 7, 12, 13, 14, 16}	E	J	K
{1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 18}	H	H	I
{1, 2, 4, 5, 6, 7, 12, 13, 14, 16, 17, 18}	I	J	K
{1, 2, 3, 4, 6, 7, 8, 9, 11, 15, 18}	J	D	E
{1, 2, 4, 5, 6, 7, 17, 18}	K	B	C

The transition diagram:



$$3. A = \varepsilon - \text{closure}(\{0\}) = \{0, 1, 3\}.$$

$$B = \delta_D(A, a) = \varepsilon - \text{closure}(\{2\}) = \{1, 2, 3\}.$$

$$C = \delta_D(A, b) = \varepsilon - \text{closure}(\{4\}) = \{4, 5, 7\}.$$

$$D = \delta_D(B, a) = \varepsilon - \text{closure}(\{2\}) = B.$$

$$E = \delta_D(B, b) = \varepsilon - \text{closure}(\{4\}) = C.$$

$$F = \delta_D(C, a) = \varepsilon - \text{closure}(\{6\}) = \{5, 6, 7\}.$$

$$G = \delta_D(C, b) = \varepsilon - \text{closure}(\{8\}) = \{8, 9, 11\}.$$

$$H = \delta_D(F, a) = \varepsilon - \text{closure}(\{6\}) = F.$$

$$I = \delta_D(F, b) = \varepsilon - \text{closure}(\{8\}) = G.$$

$$J = \delta_D(G, a) = \varepsilon - \text{closure}(\{10\}) = \{9, 10, 11\}.$$

$$K = \delta_D(G, b) = \varepsilon - \text{closure}(\{12\}) = \{12, 13, 15\}.$$

$$L = \delta_D(J, a) = \varepsilon - \text{closure}(\{10\}) = J.$$

$$M = \delta_D(J, b) = \varepsilon - \text{closure}(\{12\}) = K.$$

$$N = \delta_D(K, a) = \varepsilon - \text{closure}(\{14\}) = \{13, 14, 15\}.$$

$$O = \delta_D(K, b) = \emptyset.$$

$$P = \delta_D(N, a) = \varepsilon - \text{closure}(\{14\}) = N.$$

$$Q = \delta_D(N, b) = \emptyset = O.$$

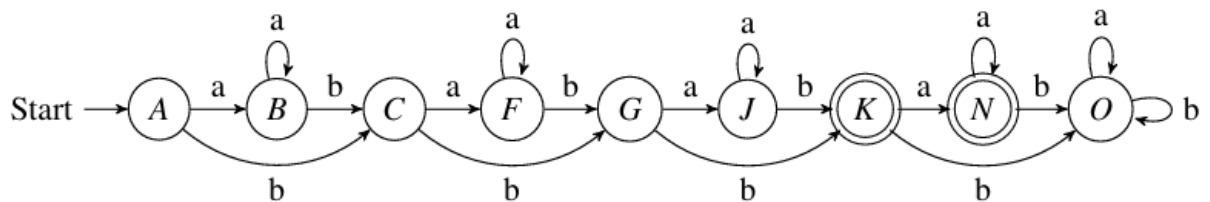
$$R = \delta_D(O, a) = \emptyset = O.$$

$$S = \delta_D(O, b) = \emptyset = O.$$

The transition table of the DFA (Strating state is A and finite states are {K, N}):

NFA STATE	DFA STATE	a	b
{0, 1, 3}	A	B	C
{1, 2, 3}	B	B	C
{4, 5, 7}	C	F	G
{5, 6, 7}	F	F	G
{8, 9, 11}	G	J	K
{9, 10, 11}	J	J	K
{12, 13, 15}	K	N	O
{13, 14, 15}	N	N	O
$\emptyset$	O	O	O

The transition diagram:

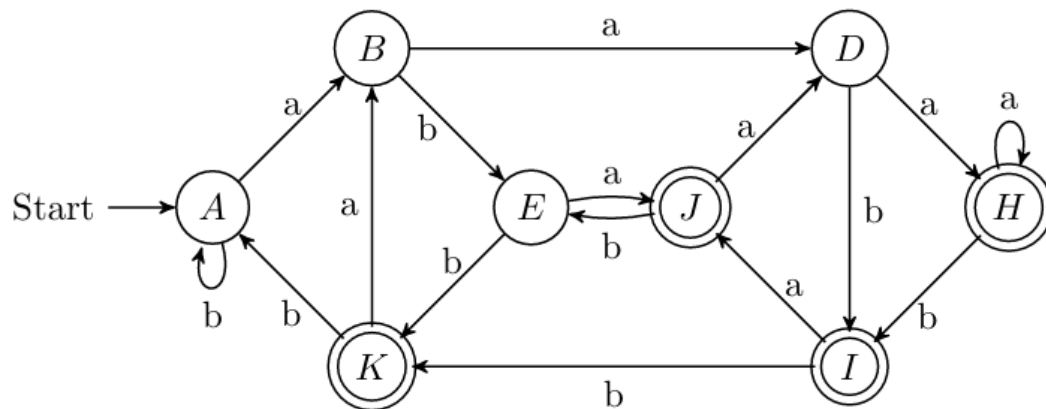


2 Optional Exercises

Exercise 1

Regular expression 2:

- (1) Initial partition: $G_1 = \{A, B, C, D, E\}, G_2 = \{H, I, J, K\}$.
- (2) Handling group G_1 : b splits it into two groups $G_3 = \{A, B, C\}, G_4 = \{D, E\}$.
- (3) Handling group G_3 : b splits it into two groups $G_5 = \{A, C\}, G_6 = \{B\}$.
- (4) Handling group G_4 : a splits it into two groups $G_7 = \{D\}, G_8 = \{E\}$.
- (5) Handling group G_2 : a splits it into two groups $G_9 = \{H, I\}, G_{10} = \{J, K\}$.
- (6) Handling group G_9 : b splits it into two groups $G_{11} = \{H\}, G_{12} = \{I\}$.
- (7) Handling group G_{10} : b splits it into two groups $G_{13} = \{J\}, G_{14} = \{K\}$.
- (8) Picking A, B, D, E, H, I, J, K as the representatives to construct the minimum-state DFA.



Regular expression 3:

- (1) Initial partition: $G_1 = \{A, B, C, F, G, J, O\}, G_2 = \{K, N\}$.
- (2) Handling group G_1 : b splits it into two groups $G_3 = \{A, B, C, F, O\}, G_4 = \{G, J\}$.
- (3) Handling group G_3 : b splits it into two groups $G_5 = \{A, B, O\}, G_6 = \{C, F\}$.
- (4) Handling group G_5 : b splits it into two groups $G_7 = \{A, B\}, G_8 = \{O\}$.
- (5) Picking A, C, G, K, O as the representatives to construct the minimum-state DFA.

